



**JOMO KENYATTA UNIVERSITY  
OF  
AGRICULTURE & TECHNOLOGY**

SCHOOL OF OPEN, DISTANCE AND eLEARNING

P.O. Box 62000, 00200

Nairobi, Kenya

E-mail: [elearning@jkuat.ac.ke](mailto:elearning@jkuat.ac.ke)

BIT 2223 Mobile Computing

LAST REVISION ON October 29, 2013





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to [elarning@jkuat.ac.ke](mailto:elarning@jkuat.ac.ke).



## LESSON 8

### Internet Protocols

#### Objectives

By the end of this topic you should be able to;

- *Understand the concept of mobile IPv4*
- *Understand the concept of mobile IPv6*
- *Differentiate between the two versions of Mobile IP*

#### 8.1. Introduction

Internet Protocol is one of the major protocol in TCP/IP protocols suite. This protocol works at Network layer of OSI model and at Internet layer of TCP/IP model. Thus this protocol has



## BIT 2223 Mobile Computing

the responsibility of identification of hosts based upon their logical addresses and to route data between/among them over the underlying network. IP provides a mechanism to uniquely identify host by IP addressing scheme. IP uses best effort delivery, i.e. it does not guarantee that packets would be delivered to destined host but it will do its best to reach the destination. Internet Protocol version 4 uses 32-bit logical address.

Mobile IP was developed for IPv4, but IPv6 simplifies the protocols. The next version of IP, IPv6, is designed to be an evolutionary step from IPv4. IPv6 addresses are 128 bits long. Mobility support in IPv6 solves many of the problems of basic Mobile IP.



## 8.2. IPv4

IP identifies each node uniquely, using an IP address that designates its physical attachment to the Internet. IP addresses are 32-bit long integers and are represented in a dotted decimal format (e.g., 128.55.44.1), for ease of use. IPv4 has proven itself as a robust routable addressing protocol and has served human being for decades on its best-effort-delivery mechanism. It was designed in early 80's and did not get any major change afterward. At the time of its birth, Internet was limited only to a few Universities for their research and to Department of Defense. IPv4 is 32 bits long which offers around 4,294,967,296 (2<sup>32</sup>) addresses. This address space was considered more than enough that time. Given below are major points which played key role in birth of IPv6:



## BIT 2223 Mobile Computing

---

- Internet has grown exponentially and the address space allowed by IPv4 is saturating. There is a requirement of protocol which can satisfy the need of future Internet addresses which are expected to grow in an unexpected manner.
- Using features such as NAT has made the Internet discontinuous i.e. one part which belongs to intranet, primarily uses private IP addresses; which has to go through number of mechanism to reach the other part, the Internet, which is on public IP addresses.
- IPv4 on its own does not provide any security feature which is vulnerable as data on Internet. Its public domain is never safe. Data has to be encrypted with some other security application before being sent on Internet.



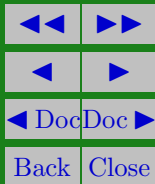
## BIT 2223 Mobile Computing

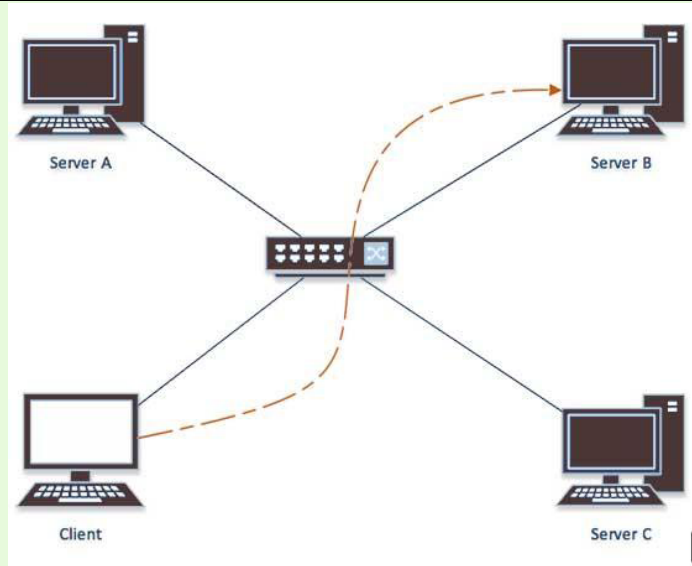
---

- Data prioritization in IPv4 is not up to date. Though IPv4 has few bits reserved for Type of Service or Quality of Service, they do not provide much functionality.
- IPv4 enabled clients can be configured manually or they need some address configuration mechanism. There exists no technique which can configure a device to have globally unique IP address.

IPv4 supports three different types of addressing modes:

1. **Unicast Addressing Mode:** In this mode, data is sent only to one destined host. The Destination Address field contains 32- bit IP address of the destination host. Here client sends data to the targeted server.





2. **Broadcast Addressing Mode:** In this mode the packet is addressed to all hosts in a network segment. The Desti-

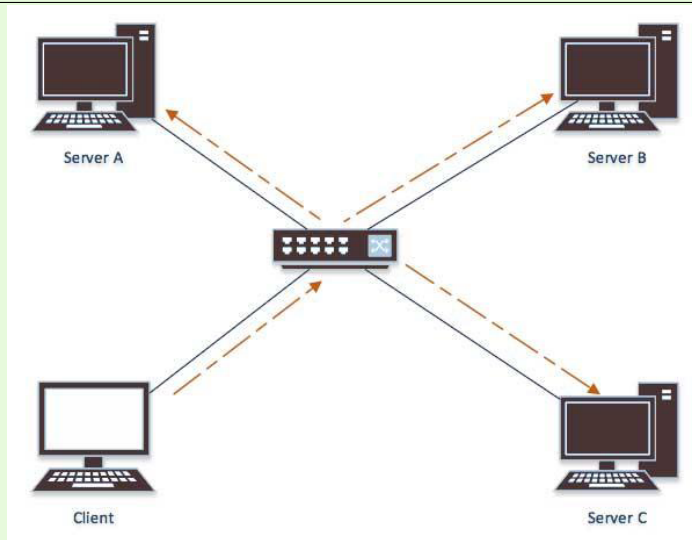




## BIT 2223 Mobile Computing

---

nation Address field contains special broadcast address i.e. 255.255.255.255. When a host sees this packet on the network, it is bound to process it. Here client sends packet, which is entertained by all the Servers.



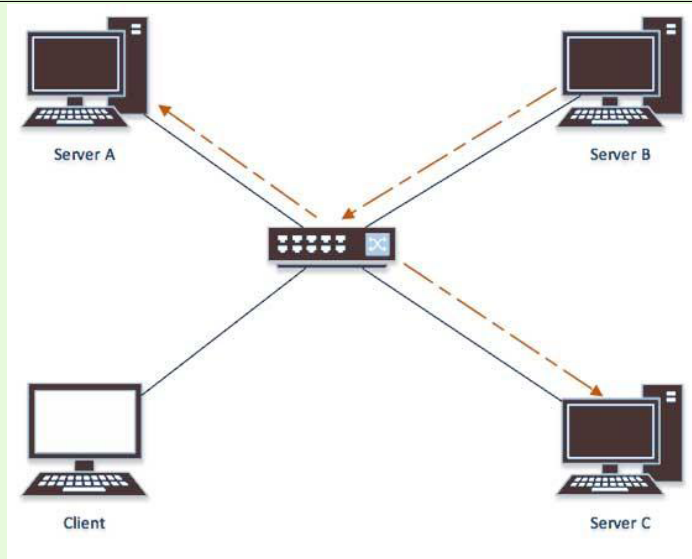
3. **Multicast Addressing Mode:** This mode is a mix of previous two modes, i.e. the packet sent is neither des-



## BIT 2223 Mobile Computing

---

limited to a single host nor all the hosts on the segment. In this packet, the Destination Address contains special address which starts with 224.x.x.x and can be entertained by more than one host.





### 8.2.1. Hierarchical Addressing Scheme

- IPv4 uses hierarchical addressing scheme. An IP address which is 32-bits in length, is divided into two or three parts as depicted:



- A single IP address can contain information about the network and its sub-network and ultimately the host. This scheme enables IP Address to be hierarchical where a network can have many sub-networks which in turn can have many hosts.



- Subnet Mask

The 32-bit IP address contains information about the host and its network. It is very necessary to distinguish both. For this, routers use Subnet Mask, which is as long as the size of the network address in the IP address. Subnet Mask is also 32 bits long. If the IP address in binary is ANDed with its Subnet Mask, the result yields the Network address. For example, say the IP Address 192.168.1.152 and the Subnet Mask is 255.255.255.0 then

|         |               |          |          |          |          |         |
|---------|---------------|----------|----------|----------|----------|---------|
| IP      | 192.168.1.152 | 11000000 | 10101000 | 00000001 | 10011000 | } ANDed |
| Mask    | 255.255.255.0 | 11111111 | 11111111 | 11111111 | 00000000 |         |
| Network | 192.168.1.0   | 11000000 | 10101000 | 00000001 | 00000000 | Result  |

This way Subnet Mask helps extract Network ID and Host from an IP Address. It can be identified now that 192.168.1.0



## BIT 2223 Mobile Computing

is the Network number and 192.168.1.152 is the host on that network.

### • Binary Representation

The positional value method is the simplest form of converting binary from decimal value. IP address is 32 bit value which is divided into 4 octets. A binary octet contains 8 bits and the value of each bit can be determined by the position of bit value '1' in the octet.

| MSB              | 8 <sup>th</sup> | 7 <sup>th</sup> | 6 <sup>th</sup> | 5 <sup>th</sup> | 4 <sup>th</sup> | 3 <sup>rd</sup> | 2 <sup>nd</sup> | 1 <sup>st</sup> | LSB |
|------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----|
|                  | 1               | 1               | 1               | 1               | 1               | 1               | 1               | 1               |     |
| Positional Value | 128             | 64              | 32              | 16              | 8               | 4               | 2               | 1               |     |

Positional value of bits is determined by 2 raised to power (position - 1), that is the value of a bit 1 at position 6 is  $2^{6-1}$



that is 25 that is 32. The total value of the octet is determined by adding up the positional value of bits. The value of 11000000 is  $128+64 = 192$ .

### 8.2.2. IPv4 Address Classes

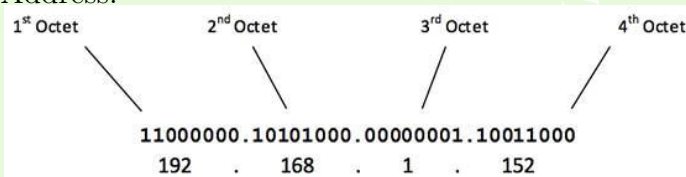
Internet Protocol hierarchy contains several classes of IP Addresses to be used efficiently in various situation as per the requirement of hosts per network. Broadly, IPv4 Addressing system is divided into 5 classes of IP Addresses. All the 5 classes are identified by the first octet of IP Address. Internet Corporation for Assigned Names and Numbers - responsible for assigning IP addresses.

The first octet referred here is the left most of all. The octets numbered as follows depicting dotted decimal notation of





IP Address:



## 8.3. IPv6

Internet Protocol version 6, is a new addressing protocol designed to incorporate whole sort of requirement of future internet known to us as Internet version 2. This protocol as its predecessor IPv4, works on Network Layer (Layer-3). Along with its offering of enormous amount of logical address space, this protocol has ample of features which addresses today's shortcoming of IPv4. IPv6 addresses are 128 bits long. Mobility support in



IPv6 solves many of the problems of basic Mobile IP.

### 8.3.1. Types of IPv6 Addresses

1. **Unicast:** Address of a single interface. One-to-one delivery to single interface
2. **Multicast:** Address of a set of interfaces. One-to-many delivery to all interfaces in the set
3. **Anycast:** Address of a set of interfaces. One-to-one-of-many delivery to a single interface in the set that is closest

#### ● IPv6 Multicast Address

IP multicast address has a prefix FF00::/8 (1111 1111); the second octet defines the lifetime and scope of the multicast address



8-bit 4-bit 4-bit 112-bit . 23 1111 1111 Lifetime Scope Group-  
ID

### 8.3.2. Advantages of Mobile IPv6 over Mobile IPv4:

- Route Optimization is built as a fundamental part of Mobile IPv6 unlike Mobile IPv4 where it is an optional set of extensions that may not be supported by all nodes.
- Foreign Agents are not needed in Mobile IPv6. The enhanced features of IPv6 like Neighbour Discovery and Address. Auto-configuration enables mobile nodes to function in any location without the services of any special router in that location.
- In Mobile IPv4, when a mobile node communicates with a correspondent node, it puts its home address as the source



## BIT 2223 Mobile Computing

address of the packet. Thus “ingress filtering routers” used to filter out the packets as the source address of the packet is different from the network from which the packet originated. This problem is tackled in Mobile IPv6 by putting the care-of address as the source address and having a Home Address Destination option, allowing the use of the care-of address to be transparent over the IP layer



## Revision Questions

**EXERCISE 1.**  Clearly describe IPV6 and state its application in mobile computing

**Example** . Compare and contrast IPv4 & IPv6 addresses.

*Solution:* For revision. □

**EXERCISE 2.**  Explain the advantages of IP V6.



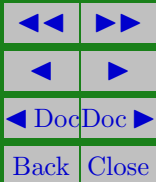
JKUAT SODeL  
©2013

BIT 2223 Mobile Computing

## Solutions to Exercises

Exercise 1. For your revision.

Exercise 1





JKUAT SODeL  
©2013

Exercise 2. For revision.

Exercise 2

